Machine learning - Project 1 report

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***Abstract*—This report presents a machine learning model designed to predict the likelihood of coronary heart disease (MICHD) based on lifestyle and clinical factors. Using data from the Behavioral Risk Factor Surveillance System (BRFSS), we developed a binary classification model using regularized logistic regression. The goal is to identify individuals who may be at risk of the disease and could benefit from early screening. However, the model's performance depends on the quality of the data and its ability to make accurate predictions.**

I. Introduction

The objective of this project is to develop a machine learning model capable of estimating the likelihood of coronary heart disease (MICHD) based on individual lifestyle and clinical factors. Early identification of individuals at elevated risk is an important component of cardiovascular disease prevention. The dataset used in this study originates from Behavioral Risk factor Surveillance System [1], a surveillance system that collects health-related information through telephone surveys. Using these data, we implemented a binary classification model based on regularized logistic regression to estimate the probability of an individual belonging to the positive class.

II. Models and Methods

A. Model

We selected logistic regression as our classification model because it is well suited for binary classification tasks. It estimates the probability of an individual belonging to a given class based on their lifestyle and clinical characteristics. It computes a weighted linear combination of the input features and transforms the result into a probability using the sigmoid function. A decision threshold is then applied to assign each observation to either the positive or negative class.

B. Algorithms

- Regularized Logistic Regression

  We used regularized logistic regression to classify observations into two categories: individuals belonging to the positive class and individuals belonging to the negative class.
  The objective function minimized during training is defined as

  $$L(w) = \frac{1}{N}\sum_{n=1}^{N} -y_n x_n^T w + \log(1 + e^{x_n^T w}) + \frac{\lambda}{2}\|w\|_2^2$$

  where N denotes the number of training observations, x_n is the feature vector of observation n, $y_n \in \{0, 1\}$ is its target label, w is the parameter vector, and λ is the regularization coefficient.

- Stochastic Gradient Descent

  We used stochastic gradient descent (SGD) to optimize the objective function. At each iteration, the parameter vector was updated as follows :

  $$w^{new} = w^{old} - \gamma \nabla L_{batch}(w^{old})$$

  where γ denotes the learning rate and L_{batch} is the loss computed on a selected batch.
  Batch size was set to one observation, which limits the computational cost of the model. This approximation is valid since the stochastic gradient descent result converges to that of the gradient descent for a high number of iterations.

- K-Fold Cross-Validation

  We used five-fold cross-validation to support hyperparameter selection. The training data are divided into five folds. In each iteration, four folds are used for training and the remaining fold is used for validation. This process is repeated until each fold has served as the validation set.

- Sigmoid function

  We use the sigmoid function

  $$p = \frac{1}{1 + e^{-xw}}$$

  once we obtain the training weights to determine the probability of the subject to get the disease, where x is the test data and w is the optimal weights vector.

C. Preprocessing

We performed one-hot encoding to address the problem of categorical features. A linear model treats categories (e.g. $1, 2, 3, 4, 5$) as numbers, implying that category 5 is "five times greater" than category 1. One-hot encoding turns them into independent binary values, allowing logistic regression weights (w) to learn a distinct coefficient for each specific category independently.
We noticed that an important number of features were defined for only a minority of data points. We started by removing the features for which 95% or more of the data were NaN.
We also removed the features that are not relevant for this

study, such as the phone number. Each feature corresponds to a question asked during the interview [2]. We computed the correlation between the different features and removed correlated features to reduce the redundancy in the dataset.
Then, we computed the median value for each remaining feature, and replaced any NaN values of this feature with the median. This way, we avoid computations with NaN in a way that should not affect the predictions of the model.
Next, we standardized the data using the Z-score normalization i.e. we substracted the mean and divided by the standard deviation for each feature. All features now have zero mean and unit variance.
Finally, we added a column of 1s before the data matrix to account for the bias weight.

D. Protocol

We started by loading the data. The dataset was split in two: the training dataset, composed of 328135 subjects, and the test dataset, composed of 109379 subjects.
Then, we did a preprocessing of both datasets with the method mentioned in Subsection II-C.
Next, we had to format the result for logistic regression. We switched from $\{-1, 1\}$ to $\{0, 1\}$ for the category assignments. Next, we set our hyperparameters. We tried different parameters and obtained the best results with 500 iterations, $\gamma = 0.3$, $\lambda = 0.0005$, and the decision threshold at 0.2. We then train our model with the regularized logistic regression, using stochastic gradient descent with the hyperparameters mentioned above.
Finally, we make the predictions on the test set : we use the sigmoid function to calculate the probability for each subject in the test set to develop the disease, using the weights that we determined with the previous step. If this probability exceeds a certain threshold (in our case 0.2), we assign the subject to 1 (i.e. will get the disease), else we assign -1.

III. Results

The results of our submissions were:

Param	Sub1	Sub2	Sub3	Sub4
F1 score	0.183	0.414	0.412	0.436
Accuracy	0.915	0.863	0.865	0.879

TABLE I
THE RESULTS OF THE MOST RELEVANT SUBMISSIONS MADE IN AI CROWD.

For each submission, we added different steps to achieve the best submission explained in the previous section. The first submission was a test of the model and the implementations and was a basic application of the required methods. In the second submission, we added preprocessing of the data and cross-validation. In the third submission, we switched from gradient descent to stochastic gradient descent. In the last submission, we tried varying the hyperparameters. One submission had a better F1 score (0.437) but we have not been able to reproduce it. The first submission had the best accuracy. However, this does not mean a good model. As we can see, the F1 score is particularly low, which is typical for this type of dataset with an imbalance. The higher number of data point for one class imbalances the whole prediction model which predict the majority of the major class, leaving the minority class not being accounted for.

IV. Discussion

The four possible outcomes of binary classification (where Positive (1) = has disease, and Negative (0) = healthy) are: - True Positives (TP): Correctly predicted positive cases.
- True Negatives (TN): Correctly predicted negative cases.
- False Positives (FP): Incorrectly predicted positive cases (Type I error).
- False Negatives (FN): Incorrectly predicted negative cases (Type II error).

To determine how well the model is doing, several metrics were considered.

- Precision

  The Precision measures how many of the positively predicted instances are actually correct.

$$Precision = \frac{TP}{TP + FP}$$

- Recall (Sensitivity / True Positive Rate)

  The Recall measures how many of the actual positive instances the model managed to detect.

$$Recall = \frac{TP}{TP + FN}$$

- F1-Score

  The F1-Score is the harmonic mean of precision and recall. It provides a balanced score that penalizes extreme imbalances between precision and recall.

$$F1\text{-}Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} = \frac{2TP}{2TP + FP + FN}$$

One strength of the approach is the ability to reach a high number of people via a non-invasive method: it is much easier to have thousands of people answer a phone survey than to have a thorough medical examination. Once the model is trained, it is also a fast approach: take the survey and get the result. No need for waiting times between two visits to the doctor.
But there are also downsides to the approach. First, it is a model and thus it is not completely accurate, especially since it relies on a probability threshold. Then, despite the high number of answers to the survey, several questions had a very low answer rate, which does not allow the study of the importance of these parameters. The answers are also given by people with no possibility to determine their accuracy.
To summarize, it is a practical method, but the results cannot be 100% accurate.

REFERENCES

[1] CDC – 2015 BRFSS survey data and documentation. [Online]. Available: https://www.cdc.gov/brfss/annual_data/annual_2015.html

[2] "2015 behavioral risk factor surveillance system questionnaire." [Online]. Available: https://www.cdc.gov/brfss/questionnaires/pdf-ques/2015-brfss-questionnaire-12-29-14.pdf